M3: Executed cover with LLM planners

10 November 2026

1 Result and implication

Task completion hides much of the damage caused by executed cover. With Qwen3:8B, live-state cover completed 102/120 requested tasks, but only 7/120 final states matched the intended business result. With Llama3.1:8B, the corresponding counts were 76/120 and 3/120. Thus 95 and 73 completed tasks still had incorrect global state. Relative to no cover, exact state integrity fell from 116 to 7 and from 69 to 3; both paired reductions survived Holm correction. Goal completion fell from 114 to 102 and from 82 to 76, with only the Qwen reduction significant after correction. These constructed adverse writes make the practical consequence measurable: an executed-cover evaluation needs a state-integrity endpoint alongside task completion and traffic inference.

The state loss bought no distinctive improvement in the measured recognition scores. With one or two randomly placed decoys, the validation-selected LR/RF attacker scored 61–63/120 across Qwen cover conditions and 72–82/120 across Llama cover conditions, compared with 118/120 and 113/120 without cover. All ten cover-versus-baseline primary comparisons survived the predeclared Holm correction. Live writes scored 62/120 and 80/120, within the range already observed for proxy-discard, read, and isolated cover. Residual exposure remained substantial: a last-destination rule reached 75–77/120 in Qwen cover conditions and exceeded the selected LR/RF attack. These are observed attack results, not equivalence or privacy upper bounds. Call-byte costs were 1.28–1.35 times the paired no-cover runs.

Aggregate scores hide variation in the visible-response control's individual execution paths. The hidden and visible isolated conditions returned identical cover responses in all 120 pairs per model. Their honest action sequences differed in 16 Qwen pairs and 65 Llama pairs. Mean prompt tokens rose by 18.1% and 18.9%, respectively. The predeclared visible-versus-hidden comparisons for recognition, goal completion, and state integrity all had Holm-adjusted $p = 1$. The paired path differences therefore describe planner behavior; they do not establish that exposing marked responses improves or degrades average utility. Hidden-arm path variation is also retained in the released audit.

2 Paired design

The experiment replaces the reference planner with fixed local Qwen3:8B and Llama3.1:8B weights. For each model, all six conditions use the same 300 synthetic ticket fixtures: 120 training, 60 validation, and 120 test fixtures. Each test cell contains 20 fixtures for each of six requested actions. The 3,600 formal episodes therefore contain 1,440 held-out executions of 120 distinct paired test fixtures. Phrasings come from the existing workload; the split holds out fixtures, not task templates or deployments.

The conditions are no cover, proxy discard, non-mutating read cover, isolated hidden execution, live-state writes, and isolated execution whose responses are visible to the planner. One or two decoys and their positions are drawn before collection. The visible condition forwards explicitly marked background results. The planner chooses checks, business actions, target order, recovery, and completion; a controller enforces prerequisite checks without selecting the correct business action.

Seven local MCP Streamable HTTP services use actual TLS 1.3/HTTP/2 sockets. Request and response bodies use 4,096- and 16,384-byte padding quanta. The observer receives service destinations and bidirectional ciphertext byte counts within supplied call and episode windows. All endpoints use the same `/mcp` path. This measurement concerns the tool link; model-link exposure, physical timing, and WAN behavior require separate observations. The proxy-discard condition synthesizes replies on a disposable fixture copy, while isolated execution mutates a persistent clone. Business services and adverse writes are controlled synthetic mechanisms.

Table 1: LLM-planned executed cover on 120 paired held-out fixtures per cell. The primary LR/RF attacker is selected on validation data. All attack columns count correct predictions out of 120; chance is 20/120. The interval conditions on the fitted attacker and synthetic fixtures.

Model	Cover	Frozen	LR	RF	Last	Penultimate	Selected BA [95% CI]
Qwen3:8B	No cover	118	118	117	116	14	0.983 [0.958, 1.000] (LR)
Qwen3:8B	Proxy discard	44	63	62	77	43	0.517 [0.425, 0.608] (RF)
Qwen3:8B	Read cover	42	59	62	77	42	0.517 [0.425, 0.608] (RF)
Qwen3:8B	Isolated, hidden	43	62	61	77	42	0.508 [0.417, 0.600] (RF)
Qwen3:8B	Live write	42	60	62	75	44	0.517 [0.425, 0.608] (RF)
Qwen3:8B	Isolated, visible	40	63	63	77	42	0.525 [0.433, 0.617] (RF)
Llama3.1:8B	No cover	113	113	110	79	64	0.942 [0.900, 0.975] (LR)
Llama3.1:8B	Proxy discard	54	81	79	67	61	0.658 [0.575, 0.733] (RF)
Llama3.1:8B	Read cover	57	78	72	67	62	0.600 [0.517, 0.683] (RF)
Llama3.1:8B	Isolated, hidden	56	81	82	67	60	0.683 [0.600, 0.758] (RF)
Llama3.1:8B	Live write	58	80	75	65	59	0.667 [0.583, 0.750] (LR)
Llama3.1:8B	Isolated, visible	58	79	81	68	64	0.675 [0.592, 0.750] (RF)

Table 2: Task outcomes and measured costs on the same 120 held-out fixtures. State integrity compares the entire business state with the single intended-action reference. Byte ratios are means of paired ratios; total TLS bytes include all seven MCP initializations, while call bytes exclude initialization.

Model	Cover	Goal	State	Goal + state error	Call bytes ×	Total bytes ×	Model calls
Qwen3:8B	No cover	114/120	116/120	0/120	1.00	1.00	4.06
Qwen3:8B	Proxy discard	114/120	116/120	0/120	1.34	1.15	4.06
Qwen3:8B	Read cover	114/120	115/120	0/120	1.34	1.15	4.07
Qwen3:8B	Isolated, hidden	114/120	115/120	0/120	1.34	1.15	4.07
Qwen3:8B	Live write	102/120	7/120	95/120	1.34	1.15	4.08
Qwen3:8B	Isolated, visible	113/120	114/120	1/120	1.35	1.15	4.08
Llama3.1:8B	No cover	82/120	69/120	13/120	1.00	1.00	5.23
Llama3.1:8B	Proxy discard	80/120	68/120	13/120	1.29	1.14	5.31
Llama3.1:8B	Read cover	84/120	71/120	14/120	1.28	1.13	5.25
Llama3.1:8B	Isolated, hidden	83/120	70/120	13/120	1.28	1.13	5.25
Llama3.1:8B	Live write	76/120	3/120	73/120	1.29	1.14	5.35
Llama3.1:8B	Isolated, visible	83/120	73/120	11/120	1.29	1.14	5.33

3 Recognition, outcomes, and costs

Recognition targets the requested action, retaining failed executions. LR and RF are fitted on training data; validation balanced accuracy chooses the primary attack per condition, with LR winning a tie. Both attacks, the baseline-trained frozen attack, and fixed position rules remain visible in Table 1. Attained accuracy measures an implemented attack, not an upper bound on available information.

Goal completion requires the intended action on the intended object, its postcondition, and all prerequisite checks. Exact state integrity compares the complete business state with the single intended-action reference. Table 2 uses the same held-out fixtures for recognition, both outcomes, and resource accounting. Call bytes exclude session setup; total TLS bytes include all seven MCP session initializations and closing.

4 Paired comparisons and planner behavior

Table 3: Selected prespecified contrasts. Differences are percentage points (condition minus reference). Each endpoint uses its own 12-comparison Holm family; the full 36 contrasts are released.

Model	Contrast	Endpoint	Difference	95% CI	Holm p
Qwen3:8B	Live–none	Goal	-10.00	[-15.00, -5.00]	0.022
Qwen3:8B	Visible–hidden	Goal	-0.83	[-4.17, 2.50]	1.000
Llama3.1:8B	Live–none	Goal	-5.00	[-10.83, 0.83]	1.000
Llama3.1:8B	Visible–hidden	Goal	0.00	[-7.50, 7.50]	1.000
Qwen3:8B	Live–none	Recognition	-46.67	[-55.83, -37.50]	3.1e-16
Qwen3:8B	Visible–hidden	Recognition	1.67	[-5.00, 8.33]	1.000
Llama3.1:8B	Live–none	Recognition	-27.50	[-35.83, -19.17]	3.1e-08
Llama3.1:8B	Visible–hidden	Recognition	-0.83	[-8.33, 6.67]	1.000
Qwen3:8B	Live–none	State	-90.83	[-95.00, -85.83]	3.7e-32
Qwen3:8B	Visible–hidden	State	-0.83	[-4.17, 2.50]	1.000
Llama3.1:8B	Live–none	State	-55.00	[-62.50, -47.50]	3e-19
Llama3.1:8B	Visible–hidden	State	2.50	[-5.00, 10.00]	1.000

All 182 planned held-out decoys triggered in every cover cell. No held-out response had invalid JSON. Qwen reached no step limits; Llama reached the limit in one no-cover, proxy-discard, read, and hidden-isolated execution, two live-write executions, and no visible-isolated executions. All remain in the denominators. The action-path audit records hidden-arm variation as well as the visible contrast. Relative to hidden isolation, proxy-discard and read had different honest choices in 1 and 0 Qwen pairs, and 15 and 15 Llama pairs; the visible arm differed in 16 and 65 pairs. These are descriptive path comparisons, not an additional significance-test family. Llama already had imperfect task execution without cover (82/120 goals and 69/120 exact states). Its no-cover traces included 69 episodes with multiple business attempts. The analysis retains that baseline behavior rather than treating every state error as caused by cover.

5 Reproduction and scope

The primary analysis fixes 12 paired contrasts per endpoint: each of five cover conditions against no cover, plus visible versus hidden isolated execution, for both models. Two-sided exact discordant-pair tests use Holm correction separately for recognition, goal completion, and state integrity. Intervals use 10,000 paired fixture resamples stratified by requested action; they condition on fitted attackers and this synthetic workload.

Collection ran on one RTX 4090 Laptop GPU with 16 GB memory, two workers, and one model loaded at a time. Per-episode wall time includes MCP setup, planner execution, journaling, and worker contention, and ends before session teardown; model loading is outside that timer. Resource counts and traffic bytes are provided separately from these local timing measurements.

The review bundle contains frozen plans, model identities, synthetic request/response and state records, ciphertext-length records, features, predictions, and offline checkers. A separately implemented replay checker checks actual decisions, decoy execution and visibility, state transitions, and TLS byte accounting. Offline checks also refit the specified attackers and reproduce validation selection, intervals, and multiplicity corrections.

The original PR replay remains a separate reference-program study. Its fixed-position and one-to-four-decoy sweep are not new LLM measurements. This extension tests constructed cover policies in the stated observer model and retains the original manuscript and reference results.